DETECH Shared Task

DEfinition and Term Extraction Challenge (DETECH)

https://detech2027.dei.unipd.it/

https://detech2026.dei.unipd.it/

https://github.com/gmdn/DETECH2026/

DETECH Shared Task

CONTEXT

- A term extraction shared task focused on biomedical texts in the domain of the gut-brain interplay (2026) and legal (2027).
- These domains presents various challenges: highly specialized and complex language, wide array of domain-specific and multiword terms.
- It comprises two subtasks:
    A. Term Extraction: identify relevant single-word and multi-word terms from English texts concerning the gut–brain interplay.
    B. Definition Generation: produce natural language definitions for the concepts designated by the extracted terms, using corpus-based evidence or automatic text generation methods.

DETECH Shared Task

TERM EXTRACTION - APPROACHES

- Standard Natural Language Toolkits: NLTK, Spacy
- Transformer-based TE: BERT models pre-trained on Italian
- LLM-based TE: zero-shot TE, LORA finetuning

DEFINITION GENERATION - APPROACHES

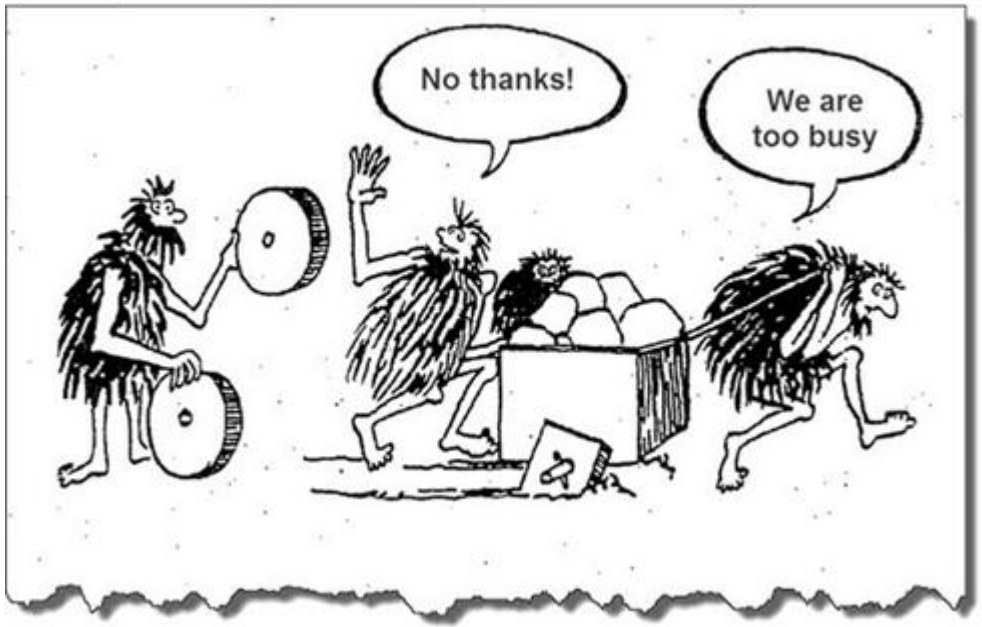
- To be defined

How to approach a shared task?

- Understand data and evaluation metrics
- Check and reproduce baseline(s), if available.
- Set up vanilla system and simpler systems
- Assess systems: define research questions/directions and verify hypotheses
- (Try to) Improve systems aligned with your research directions

How to approach a shared task?

It is always good practice to review the scientific literature to understand what has already been explored and what results have been achieved.



What can be improved?

Some ideas:

- Ensembling strategies: union, intersection, majority voting
- Second-stage filtering: generate predictions with a model, use a second model to process these predictions and filter false positives (precision-oriented strategy)
- In BERT-based solutions, use more sophisticated classification heads (e.g., CRF, LSTM, BiLSTM…)
- Improve the prompting of LLMs

What can be improved?

!! IMPORTANT !!

When working on a shared task, **your research questions and methodology** matter more than the results themselves. Start with a clear research question(s), then design experiments to address it, not the other way around.

What can be improved?

GOLD RULES

- Define your research questions clearly. What do you want to investigate? What hypotheses do you aim to test?
- Design your experimental settings accordingly. Specify what you want to measure, how you will measure it, and why your chosen methodology is appropriate for answering your research questions.
- Choose meaningful baselines and evaluation metrics. Ensure that your experiments allow for fair comparisons and that the selected metrics capture what you actually want to assess.
- Justify your experimental choices. Be able to explain why you selected specific datasets, methods, parameters, and evaluation procedures.
- Analyze and interpret your results critically. Don't just report numbers. Explain what they mean, whether they support your hypotheses, and what conclusions can reasonably be drawn from them.
- Consider negative or unexpected results. Experiments that do not confirm your hypotheses can still provide valuable insights. Avoid modifying your research questions retrospectively just to fit the results.
- Ensure reproducibility. Document your experimental setup, configurations, and procedures so that others can reproduce and validate your findings.